

National University of Singapore
School of Computing
CS1010X: Programming Methodology
Semester II, 2018/2019

Mission 8
3-station IPPT

Release date: 09 March 2019

Due: 14 April 2019, 23:59

Required Files

- mission08-template.py
- ippt.py
- pushup.csv
- situp.csv
- run.csv

Background

Every year, able-bodied Singaporean males who are in or have completed compulsory military training (a.k.a. National Service) have to take the Individual Physical Proficiency Test (IPPT). The IPPT is a physical fitness test meant to assess the fitness of our soldiers to ensure that our soldiers are physically fit in times of uncertainty.

In 2014, the Ministry of Defence announced a new test format and scoring system for IPPT, which was implemented in 2015. The new IPPT consists of the following 3 stations:

- **Push-up**, which measures upper body strength
- **Sit-up**, which measures abdominal strength
- **2.4km Run**, which measures cardiovascular fitness and lower body strength

Soldiers participating in the 3-station IPPT will first attempt as many push-ups and sit-ups that they can do in one minute each. Depending on their age, they are given points based on the number of repetitions they have completed. Next, they have to complete a 2.4km run, and would be given points depending on how fast they complete the run.

The final IPPT score is the sum of all the scores obtained at the 3 stations, which determines if they have passed or failed the IPPT. To reward and motivate soldiers to maintain a high level of fitness, a monetary reward is given for soldiers who have demonstrated a high level of physical fitness. The reward amount depends on the final score obtained, and can be broken down into Pass, Pass with Incentive, Silver, and Gold. This is known as the IPPT Award.

In this mission, you will create an IPPT calculator, which would help our soldiers calculate their IPPT score and award, given their age, and score they attained at the various stations.

This mission consists of **5** tasks, and a bonus task.

IPPT Data

The IPPT scoring criteria is provided to you in 3 different files:

- Push-ups: pushup.csv
- Sit-ups: situp.csv
- 2.4km Run: run.csv

Push-up / Sit-up

Push-ups and sit-ups in the IPPT are scored in terms of the number of repetitions that a soldier can do in one minute. pushup.csv and situp.csv lists the scores that a soldier of a certain age will obtain if they managed a particular number of repetitions.

The first row describes the number of repetitions, whereas the first column lists the age of the soldier.

From the file situp.csv, we observe that an 18-year-old who did 10 sit-ups would get 0 points (green box), whereas a 40-year-old who did 10 sit-ups would get 3 points (yellow box).

	A	B	C	D	E	F	G	H	I	J	K	L	M
1	AGE/REP	1	2	3	4	5	6	7	8	9	10	11	12
2	18	0	0	0	0	0	0	0	0	0	0	0	0
3	19	0	0	0	0	0	0	0	0	0	0	0	0
4	20	0	0	0	0	0	0	0	0	0	0	0	0
5	21	0	0	0	0	0	0	0	0	0	0	0	0
6	22	0	0	0	0	0	0	0	0	0	0	0	0
7	23	0	0	0	0	0	0	0	0	0	0	0	0
8	24	0	0	0	0	0	0	0	0	0	0	0	0
9	25	0	0	0	0	0	0	0	0	0	0	0	0
10	26	0	0	0	0	0	0	0	0	0	0	0	0
11	27	0	0	0	0	0	0	0	0	0	0	0	0
12	28	0	0	0	0	0	0	0	0	0	0	0	1
13	29	0	0	0	0	0	0	0	0	0	0	0	1
14	30	0	0	0	0	0	0	0	0	0	0	0	1
15	31	0	0	0	0	0	0	0	0	0	0	1	2
16	32	0	0	0	0	0	0	0	0	0	0	1	2
17	33	0	0	0	0	0	0	0	0	0	0	1	2
18	34	0	0	0	0	0	0	0	0	0	1	2	3
19	35	0	0	0	0	0	0	0	0	0	1	2	3
20	36	0	0	0	0	0	0	0	0	0	1	2	3
21	37	0	0	0	0	0	0	0	0	1	2	3	4
22	38	0	0	0	0	0	0	0	0	1	2	3	4
23	39	0	0	0	0	0	0	0	0	1	2	3	4
24	40	0	0	0	0	0	0	0	1	2	3	4	5
25	41	0	0	0	0	0	0	0	1	2	3	4	5
26	42	0	0	0	0	0	0	0	1	2	3	4	5
27	43	0	0	0	0	0	0	1	2	3	4	5	6

2.4km Run

The 2.4km run in the IPPT is scored based on the timing a soldier takes to complete a 2.4km run. To simplify things, we will represent the time of the run in seconds. run.csv lists the score that a soldier of a certain age will obtain if they complete the run under a particular timing.

The first row describes the time in seconds, while the first column lists the age of the soldier.

A different score is given whenever there is a timing difference of 10 seconds. From the run.csv file, we observe that an 18-year-old who ran 08:30 (510 seconds) or faster would get 50 points and one who ran between 08:31 (511 seconds) and 08:40 (520 seconds) inclusive would get 49 points.

The data file shows the scores for the timing between 510 seconds to 1100 seconds. Naturally, a timing less than 510 seconds would get 50 points, and a timing exceeding 1100 seconds would get 0 points.

Administrivia

The following functions have been provided to you:

- `make_ippt_table(pushup_table, situp_table, run_table)`
- `get_situp_table(ippt_table)`
- `get_pushup_table(ippt_table)`
- `get_run_table(ippt_table)`

Table Data Structure

To help you manipulate and access the data in a tabular fashion, we have provided an implementation of a table data structure which would help you access the scores in the IPPT table.

The functions that are provided for the table data structure:

- `create_table(data, row_keys, col_keys)`
- `access_cell(table, row_key, col_key)`

`create_table` takes in 3 parameters. The first parameter `data` is a tuple of tuples, which contains the table of data. The second parameter `row_keys` is the tuple of keys associated to each row, and the third parameter `col_keys` is the tuple of keys associated to each column.

As an example, consider the following table:

S/N	Name	Gender	Age	Course
1	Wai Hon	M	23	Business Analytics
2	Yang Shun	M	25	Computer Science
3	Xiangxin	F	18	Computer Science
4	Soedar	M	25	Computer Science

The following code sample illustrates how to represent the above data in Python using our own table data structure:

```

user_data = (("Wai Hon", "M", 23, "Business Analytics"),
             ("Yang Shun", "M", 25, "Computer Science"),
             ("Xiangxin", "F", 18, "Computer Science"),
             ("Soedar", "M", 25, "Computer Science"))

user_table = create_table(user_data,
                          (1, 2, 3, 4),
                          ("Name", "Gender", "Age", "Course"))

```

`access_cell`, is a general accessor which would retrieve a particular cell from a table given a `row_key` and a `column_key`. Compare the returned value of the execution of the `access_cell` functions below with the table above.

```

access_cell(user_table, 1, "Course")    # Business Analytics
access_cell(user_table, 2, "Age")       # 25
access_cell(user_table, 3, "Name")      # Xiangxin
access_cell(user_table, 4, "Gender")    # M

```

Task 1: Read Data (4 marks)

Implement `read_data`, a function that would read the input data file, and return a table of scores for a particular station.

```

situp_table = read_data("situp.csv")
pushup_table = read_data("pushup.csv")
run_table = read_data("run.csv")

# Sit-up score of a 24-year-old who did 10 sit-ups.
access_cell(situp_table, 24, 10)    # 0

# Push-up score of a 18-year-old who did 30 push-ups.
access_cell(pushup_table, 18, 30)    # 16

# Run score of a 30-year-old who ran 12 minutes (720 seconds)
access_cell(run_table, 30, 720)      # 36

# Since our run.csv file does not have data for 725 seconds, we should
# get None if we try to access that cell.
access_cell(run_table, 30, 725)      # None

```

Task 2: Custom Accessors (6 marks)

The default accessor function, `access_cell`, is a general accessor which retrieves a particular cell given a `row_key` and a `column_key`. As such, it is not aware of the type of data the table contained, and would not return the correct score if the `column_key` does not exist in our data.

We would like to create a custom accessor for each station. When the `column_key` does not exist, the cell with the closest `column_key` value will be returned. For example,

`pushup_score(pushup_table, 18, 61)` returns 25 which is the score in cell (18, 60) because 61 doesn't exist and the closest column is 60.

Both push-up and sit-up tables have column keys ranging from 1 to 60. The run table has column keys ranging from 510 to 1100. Moreover, the score data are in an interval of 10 seconds, we are not able to access the run score for a timing that falls within the 10 seconds interval. For example, a 30-year-old male who has a run timing of 735 seconds will be considered under the band of 740 seconds and will get 35 points (timing is rounded up to the nearest band). Using a custom accessor would allow us to retrieve the correct score.

Create new accessors for each of the stations. You may assume that the inputs are guaranteed to be non-negative integers and that all given `row_key` will be in the valid range.

Sit-up and Push-up

```
situp_score(situp_table, 24, 0)      # 0
pushup_score(pushup_table, 18, 61)    # 25
pushup_score(pushup_table, 18, 70)    # 25
```

2.4km Run

```
run_score(run_table, 30, 720)         # 36
run_score(run_table, 30, 725)         # 35
run_score(run_table, 30, 735)         # 35
```

Task 3: Calculate IPPT Award (2 marks)

An IPPT Award is awarded based on the total score that a soldier has obtained for all the 3 stations. We will use the following table to determine the IPPT Award.

Award Name	Award String	Total Score
Fail	"F"	< 51
Pass	"P"	≥ 51
Pass with Incentive	"P\$"	≥ 61
Silver	"S"	≥ 75
Gold	"G"	≥ 85

Define a function, `ippt_award`, which will return the Award String for a given IPPT Score

```
ippt_award(50)      # F
ippt_award(51)      # P
ippt_award(61)      # P$
ippt_award(75)      # S
ippt_award(85)      # G
```

Task 4: Calculate IPPT Results (2 marks)

We can now calculate the IPPT score and the qualifying award of a soldier's test attempt. The result consists of both the total points and the award, and is represented as a tuple of (Total IPPT Score, IPPT Award String).

Implement the function `ippt_results(ippt_table, age, pushup, situp, run)`, which will calculate the total IPPT score and award given the age of the soldier, number of sit-ups, number of push-ups, and 2.4km run timing.

```
ippt_results(ippt_table, 25, 30, 25, 820)      # (53, 'P')
ippt_results(ippt_table, 28, 56, 60, 530)      # (99, 'G')
ippt_results(ippt_table, 38, 18, 16, 950)      # (36, 'F')
ippt_results(ippt_table, 25, 34, 35, 817)      # (61, 'P$')
ippt_results(ippt_table, 60, 70, 65, 450)      # (100, 'G')
```

Task 5: Training Hard (4 marks)

Wai Hon's IPPT is coming soon, and he would like to determine the best training program that he should embark on in order to obtain the highest possible IPPT score before the actual IPPT. A training program is characterized by the improvement that one can obtain in each of the IPPT stations, namely:

- `rate_pushup`: Number of days it will take to increase the push-up count by 1
- `rate_situp`: Number of days it will take to increase the sit-up count by 1
- `rate_run`: Number of days it will take to decrease 2.4km run timing by 1 second

By following a training program, improvements will be made in **all** 3 stations. For instance, if `rate_pushup` is 1, `rate_situp` is 2, and `rate_run` is 3, after 10 days of following the training program, Wai Hon will improve his push-up count by 10, sit-up count by 5, and his 2.4km run timing by 3 seconds.

Implement the function `make_training_program(rate_pushup, rate_situp, rate_run)`, which returns a new function that accepts 6 parameters, `ippt_table`, `age`, `pushup`, `situp`, `run`, and `days`.

The returned function will then return the number of push-ups, sit-ups, and the 2.4km run timing that can be achieved at the end of the specified number of days, together with the improved IPPT result.

```
tp = make_training_program(7, 3, 10)
tp(ippt_table, 25, 30, 25, 820, 30)      # (34, 35, 817, (61, 'P$'))
```

Bonus: Skip Leg Day (2 bonus marks)

In Task 5, by following the training program, we are able to gain improvements in all the stations daily. We will tweak the question slightly for this bonus task.

Suppose that we are not able to gain daily improvements for all the stations, and that we have to focus on improving our score for each station one at a time. What is the best possible IPPT score we can obtain at the end of a specified number of days?

You should assume that you can improve on another station only after you have gained at least one unit of improvement in a particular station.

Hint: Have you solved a similar problem before? :)

Note: Depending on your implementation, you might get a different number of sit-up, push-up, and 2.4km run timing. However, the IPPT score and award should be the same as the sample output.

Note: The full score is capped at 18 marks for this mission.

```
tp_bonus = make_tp_bonus(7, 3, 10)
tp_bonus(ippt_table, 25, 20, 30, 800, 30)      # (20, 40, 800, (58, 'P'))
tp_bonus(ippt_table, 25, 20, 30, 800, 2)       # (20, 30, 800, (52, 'P'))
```